2 1000 randomly generated integers between 1 to 10000 (inclusive) are stored in a text file `TASK2_DATA.txt`.

Task 2.1

Write a function `read_from_file(filename)` that:

- takes the name of the text file as input argument,
- reads the data from the file, and
- returns the data as a list of integers.

Call your function `read_from_file` with `TASK2_DATA.txt`, printing the returned list and its length, using the following statements:

```
list1 = read_from_file("TASK2_DATA.txt")
print(list1)
print(len(list1))
```

[4]

Task 2.2

Write a function `quick_sort(list_of_integers)` that:

- takes a list of integers,
- implements a quick sort algorithm, and
- returns the sorted list of integers.

Call your function `quick_sort` with the data of the file `TASK2_DATA.txt`, printing the returned list, using the following statements:

```
list1 = read_from_file("TASK2_DATA.txt")
list1_sorted = quick_sort(list1)
print(list1_sorted)
```

[6]

Task 2.3

Write a function `linear_search(list_of_integers, target)` that:

- takes a **sorted** list of integers,
- implements the linear search algorithm to search for `target`, without making any redundant comparison in the search algorithm, and
- returns `True` if `target` is found, and `False` otherwise.

Call your function `linear_search` by passing in a sorted list of integers. Assign `2022` to `target`, printing the returned list. Use the following statements:

```
list1 = read_from_file("TASK2_DATA.txt")
list1_sorted = quick_sort(list1)
print(linear_search(list1_sorted, 2022))
```

[4]

Task 2.4

Write a function `binary_search(sorted_list, target)` that:

- takes a **sorted** list of integers
- implements the binary search algorithm to search for target
- returns `True` if target is found, else return `False`

Call your function `binary_search` by passing in a sorted list of integers sorted using the `quick_sort` function from **Task 2.2**, and target is 2022, printing the returned list. For example, using the following statements:

```
list1 = read_from_file('TASK2_DATA.txt')
list1_sorted = quick_sort(list1)
print(binary_search(list1_sorted, 2022))
```

[5]

Download your program code for Task 2 as

`TASK2_<your class>_<your name>.ipynb`